Lecture 2
Classification of Medical Devices in Healthcare

Regulation, Risk, Triage, and Clinical Impact

Lesson overflow

- Defines medical devices and their FDA / MDR classifications.
- Covers diagnostic, therapeutic, and preventive technologies.
- Explains regulatory risk categories (high-risk vs non-critical).
- Shows how devices are used across clinical settings , clinics, hospitals, and home care.

FDA Definition of a Medical Device

An instrument, apparatus, implement, machine, contrivance, implant, in-vitro reagent, or other similar or related article, including a component part, or accessory which is:

- (A) recognized in the official National Formulary, or the United States Pharmacopoeia, or any supplement to them,
- (B) intended for use in the diagnosis of disease or other conditions, or in the cure, mitigation, treatment, or prevention of disease, in man or other animals, or
- (C) intended to affect the structure or any function of the body of man or other animals, and which does not achieve its primary intended purposes through chemical action within or on the body of man or other animals and which is not dependent upon being metabolized for the achievement of its primary intended purposes.

The term "device" does not include software functions excluded pursuant to section 520.

* However, in some regulatory pathways and scenarios, software is considered a medical device

FDA Ref

- https://www.fda.gov/medical-devices/classify-your-medical-device/how-determine-if-your-product-medical-device



MDR Definition of a Medical Device

‘**Medical device**’ means any instrument, apparatus, appliance, software, implant, reagent, material or other article intended by the manufacturer to be used, alone or in combination, for human beings for one or more of the following specific medical purposes:

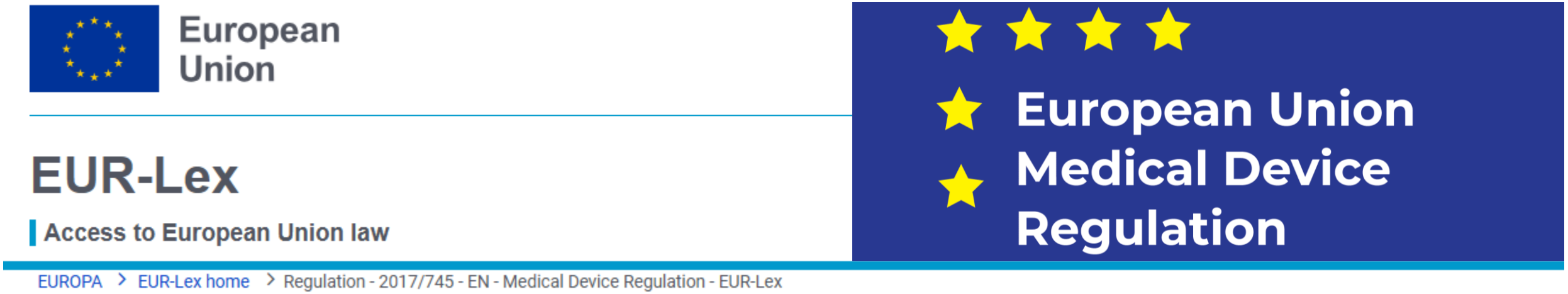
- diagnosis, prevention, monitoring, prediction, prognosis, treatment or alleviation of disease,
- diagnosis, monitoring, treatment, alleviation of, or compensation for, an injury or disability,
- investigation, replacement or modification of the anatomy or of a physiological or pathological process or state,
- providing information by means of in vitro examination of specimens derived from the human body, including organ, blood and tissue donations,

and which does not achieve its principal intended action by pharmacological, immunological or metabolic means, in or on the human body, but which may be assisted in its function by such means. The following products shall also be deemed to be medical devices:

- devices for the control or support of conception;
- products specifically intended for the cleaning, disinfection or sterilization of devices as referred to in Article 1(4) and of those referred to in the first paragraph of this point.

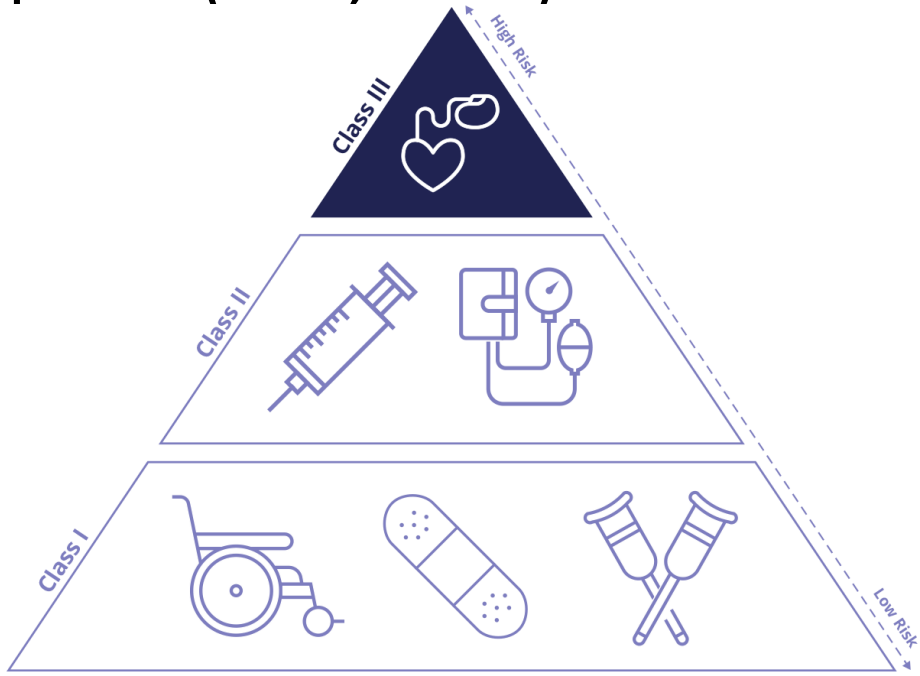
MDR Ref

- https://eur-lex.europa.eu/eli/reg/2017/745/oj/eng



FDA Device Classification (Risk-Based System)

- FDA established three regulatory classes depending on level of control needed for safety and effectiveness
- Classification depends on intended use and indications for use
- Determines type of submission/application required (FDA, 2018)
  - Class I
  - Class II
  - Class III



FDA Class I – Lowest Risk

- Not intended to sustain/support life
- Present lowest risk of injury
- Examples: elastic bandages, manual stethoscopes, examination gloves, handheld surgical instruments









FDA Class II – Moderate Risk

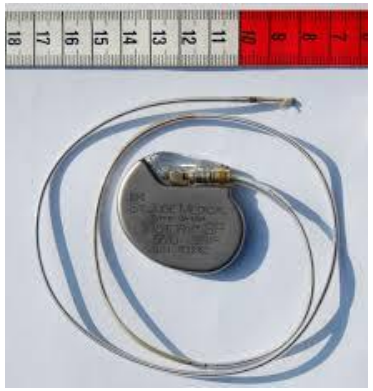
- Higher chance of patient contact and possible harm
- Require greater regulatory control
- Examples: infusion pumps, powered wheelchairs





FDA Class III – Highest Risk

- Sustain or support life, implanted in the body, or pose potential unreasonable risk
- Require premarket approval with strongest evidence of safety/effectiveness
- Examples: pacemakers, automated external defibrillators

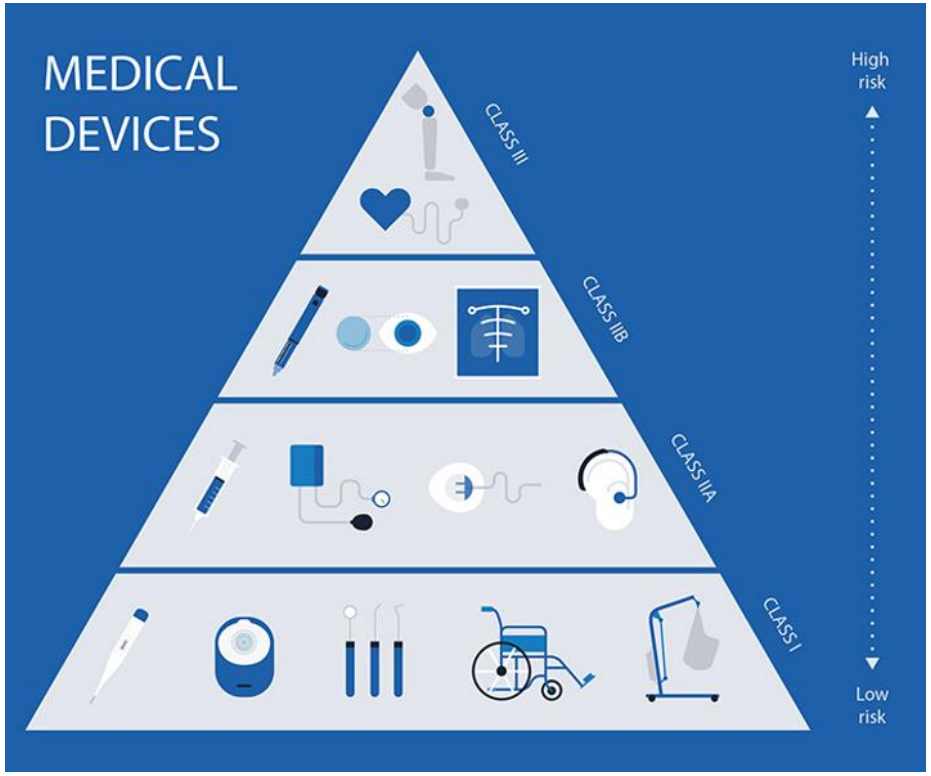




MDR Classification (European Union)

- Regulation: EU MDR 2017/745
- Devices classified into four classes based on risk and invasiveness:
  - Class I
  - Class IIa
  - Class IIb
  - Class III
- Related article:

https://openregulatory.com/articles/eu-mdr-classification-examples



MDR Classification (European Union) – Class I

- Class I – Low risk, non-invasive, non-critical use
- Examples: bandages, non-powered wheelchairs





MDR Classification (European Union) – Class IIa

- Class IIa – Medium risk, short-term invasive or diagnostic use
- Examples: dental fillings, hypodermic needles, hearing aids







MDR Classification (European Union) – Class IIb

- Class IIb – Medium-to-high risk, longer-term invasive or active devices
- Examples: ventilators, infusion pumps, anesthesia machines

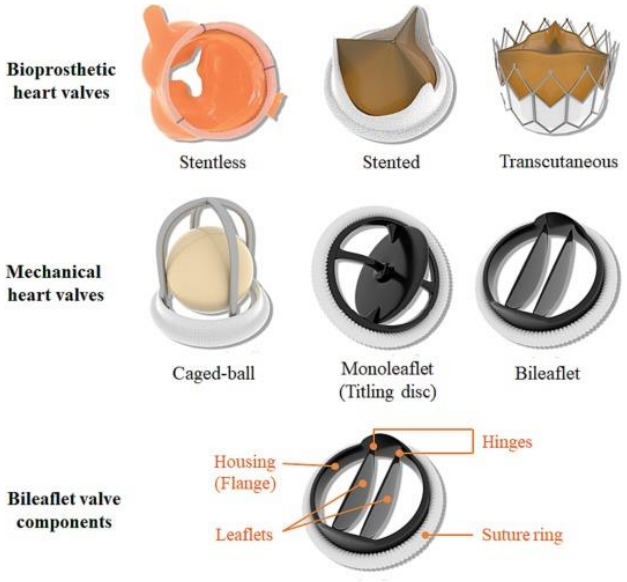




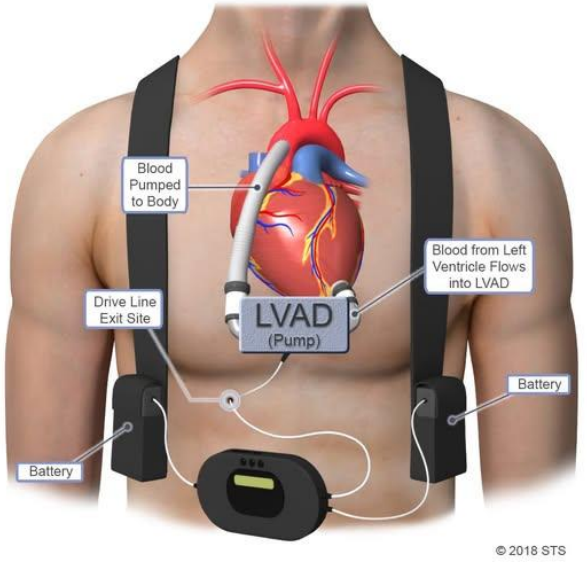


MDR Classification (European Union) – Class III

- Class III – Highest risk, sustain life, critical implants, or very invasive
- Examples: heart valves, pacemakers, implantable LVADs







Key difference MDR vs FDA

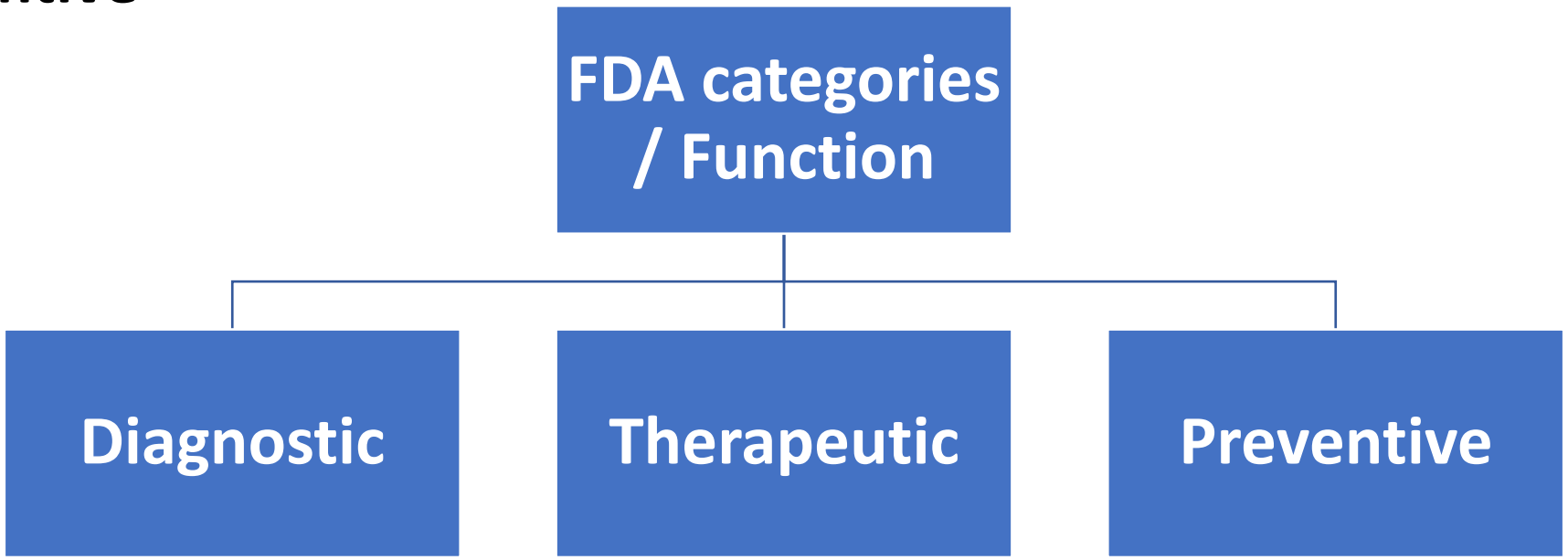
- CE MDR
  1. uses four-tier risk system (I, IIa, IIb, III)
  2. Stronger requirements for clinical evidence and post-market surveillance





General Types of Medical Technology (FDA Categorization)

- FDA categories based on device function: **Diagnostic, Therapeutic, Preventive**



FDA categories based on device function: Diagnostic Devices

- Gather physiologic or anatomic data to identify illness, injury, or disease
- Range from simple → complex:
  - **Simple**: thermometers, otoscopes (used for diagnosing infections)
  - **Complex**: physiologic monitors (ECG rhythms for atrial fibrillation)
    - Laboratory analyzers: blood tests for counts or chemical markers
    - Imaging devices:
      - X-ray (bones, teeth, lungs)
      - MRI & Ultrasound (soft tissues, heart valve motion, non-radiation)





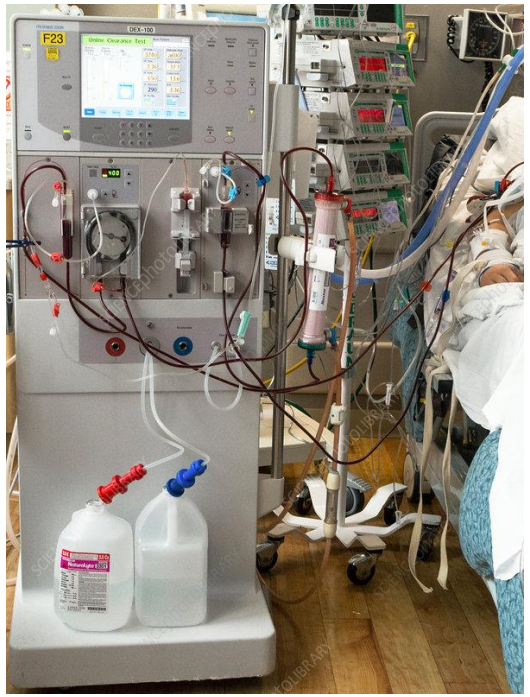


FDA categories based on device function:

Therapeutic Devices



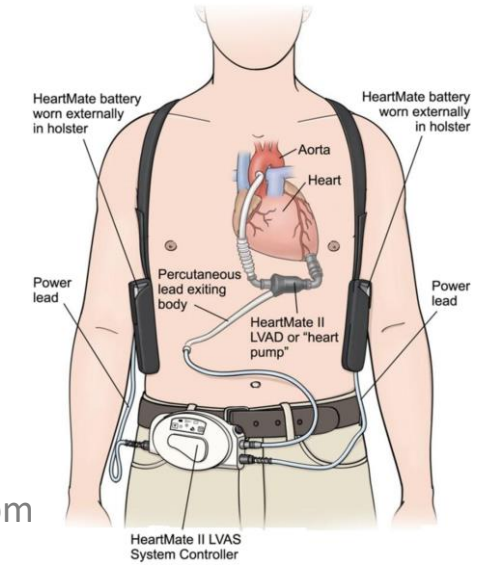
- Treat underlying disease/condition directly
- Examples:
  - Defibrillator → electric shock to restore heart rhythm
  - Hemodialysis machine → filters blood, removes toxins
  - Ventilators → support respiratory function





FDA categories based on device function:
Preventive Devices

- Aim to prevent disease or manage chronic conditions before complications
- Examples:
  - Insulin pump → maintains controlled drug concentration in body
  - LVAD (Left Ventricular Assist Device) → supports failing heart as a bridge to transplant





Regulatory Definitions & Categorization (Beyond FDA)

- FDA definitions/classifications guide manufacturers in determining submissions for market approval.
- However, in clinical engineering practice, FDA classes are NOT commonly used.
- Instead, hospital-based departments follow US healthcare regulatory bodies such as:
  - Centers for Medicare and Medicaid Services (CMS)
  - The Joint Commission

  These organizations regulate hospitals and focus on risk categorization based on potential harm to patients or caregivers if equipment fails

High-Risk (Critical) vs. Non-High-Risk (Non-Critical) Equipment

High-Risk (Critical) Equipment

- Failure may cause **serious injury or death** to patients or staff.
- Includes **life-support equipment** such as:
  - Anesthesia machines
  - Ventilators
- “High-risk” term introduced in Joint Commission standards; considered equivalent to CMS term **“critical equipment.”**
- Defined in the **Joint Commission glossary (2018)**.

High-Risk (Critical) vs. Non-High-Risk (Non-Critical) Equipment

Non-High-Risk (Non-Critical) Equipment

- Failure likely results in only **minor injury** to patients or caregivers.
- Examples:
  - Examination tables
  - Blood pressure devices
  - Physiologic monitors
- Equivalent to CMS term **“non-critical equipment.”**
- Referenced in **National Fire Protection Association (2018)**.

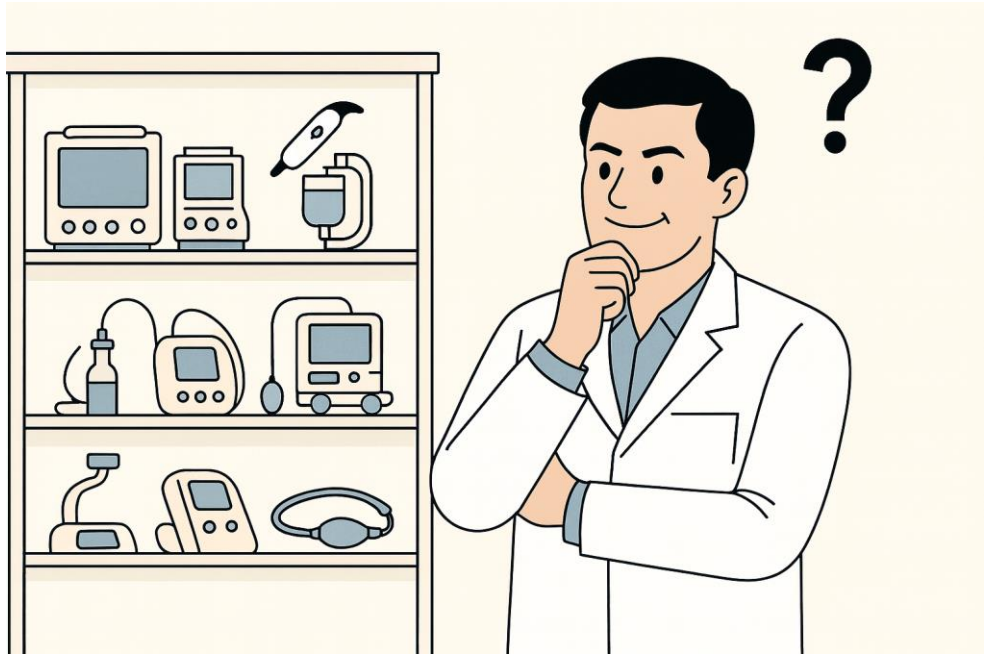
Alignment of FDA and Hospital Regulatory Categories

- FDA Classes (I, II, III) focus on regulatory submission and manufacturing control.
- CMS / Joint Commission Categories focus on operational safety and clinical use:
  - High-Risk (Critical) ≈ FDA Class IIb/III (life-supporting or sustaining devices)
  - Non-High-Risk (Non-Critical) ≈ FDA Class I/IIa (low to moderate risk devices)

  Although not identical, the two systems roughly align in how they rank patient safety risk.

Custom Medical Device Classification in Hospitals

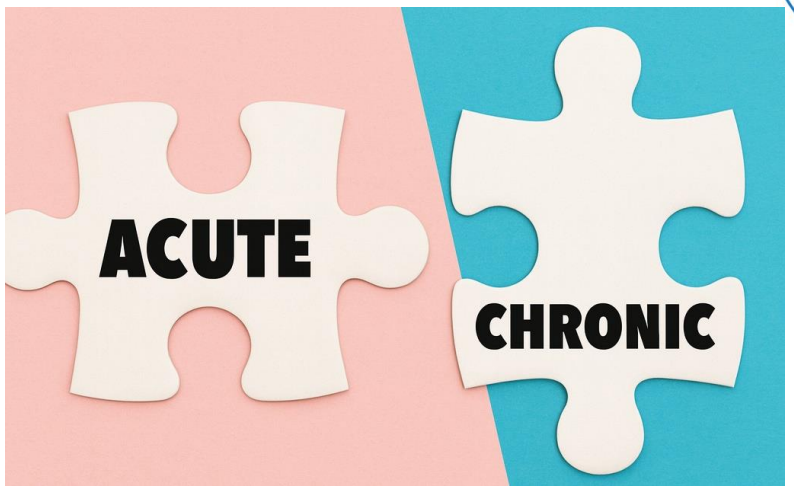
- You can create your own specific classification for your hospital or institution according to your needs, but generally you have to consider the hazard level and complexity.



Evolution of Devices in Patient Care

- Medical treatment delivery has evolved significantly since the 1800s.From simple tools → advanced medical devices.
- By the 1980s, technology became a critical component of diagnosis and treatment.
- Expansion of devices driven by:
  - Risk-averse medical environment
- Clinicians relying on technology to avoid errors or negligence.
- Result: shift in focus described as:
  - From "care of patients" → to "care of technology."
- Understanding this framework is essential for clinical engineers in modern healthcare.

Categories of Medical Care: Acute vs. Chronic



- **Acute Care**
  - Associated with illness or injury.
  - Treatment period is limited.
  - Goal: recovery and return to wellness.
- **Chronic Care**
  - Associated with long-term or permanent conditions.
  - Requires extended or lifelong care.
  - Goal is not always recovery, but **supporting quality of life**.
- **Technology Role**
  - Strongly integrated in **both acute and chronic care**.
- **Majority of medical devices** are used in **acute treatment**.


INPATIENTS
OUTPATIENTS

Patient Status in the Healthcare System

- Patients classified as **Inpatient** or **Outpatient**.
- Misconception: overnight stay automatically = inpatient.
- True definition (CMS, 2018):
  - **Inpatient** = patient is **formally admitted** with a **physician's order**.
  - **Outpatient** = receiving services without a physician's admission order.
    - Emergency department visits
    - Observation status
    - Outpatient surgery
    - Laboratory tests, x-ray imaging, other services

Inpatients VS Outpatients

- **Formal :**A patient is considered an *inpatient* only when a physician issues a formal admission order; without this order, the patient remains *outpatient*, even if they stay overnight or undergo extensive treatment.

- **General :** Inpatients stay in the hospital for ongoing monitored care, while outpatients come in for tests or procedures and go home the same day,
- but remember, time alone does not determine the classification.

Example Scenario 1: Outpatient Surgery/Procedure

- Patient scheduled for biopsy in endoscopy department.
- Excessive bleeding occurred → patient stayed in recovery for hours of observation.
- No physician admission order was written.
- Entire encounter = Outpatient event.

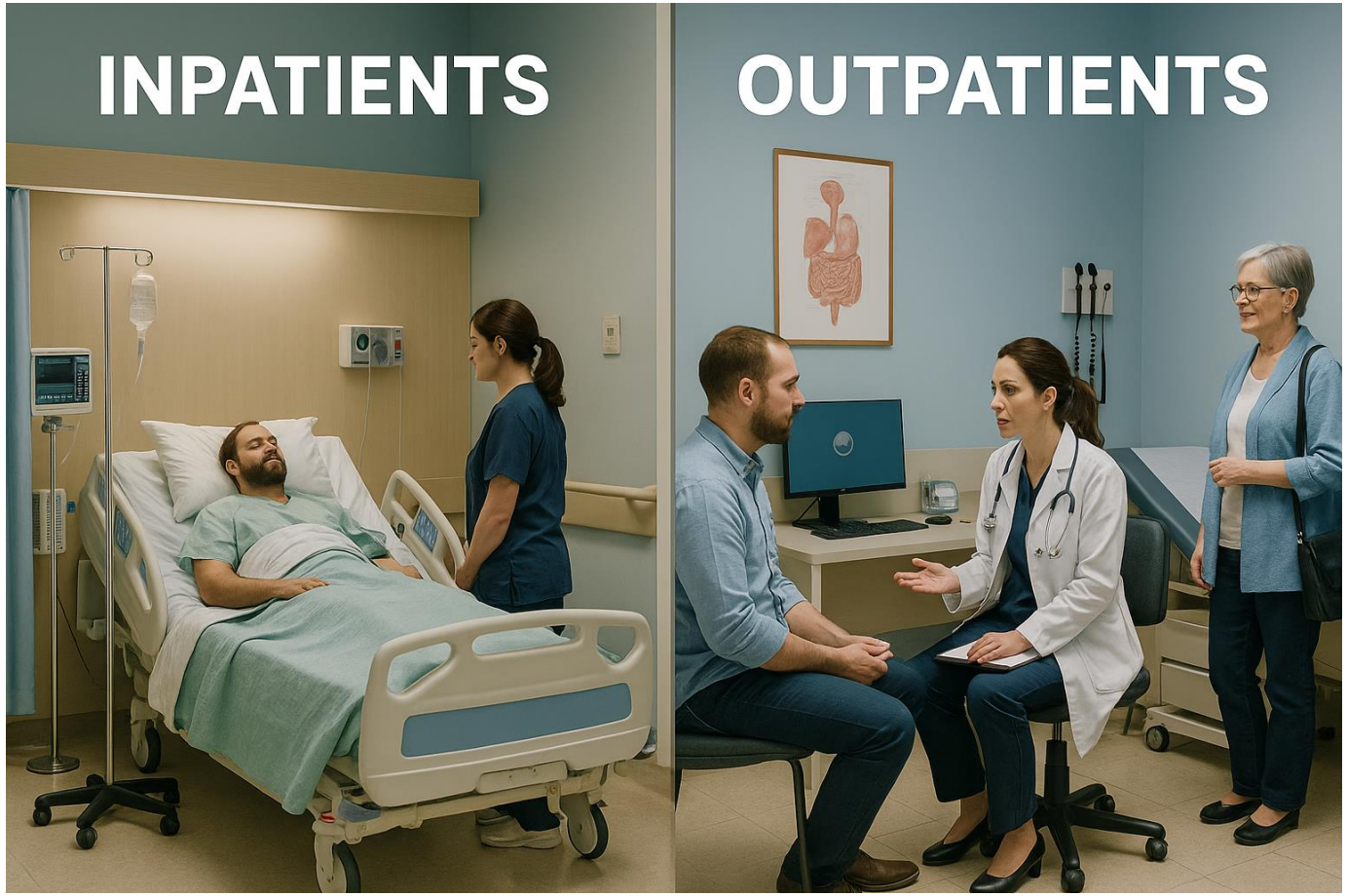
Example Scenario 2: Clinic Visit → ED → Observation → Surgery

- Patient had a stress test (outpatient clinic).Technician noted cardiac issues → referred to physician.
- Physician sent patient to emergency department for tests.
- Additional imaging + lab tests performed → still outpatient (no admission order).Physician placed patient in observation status (could last 23 hrs. to several days).Patient stayed overnight in hospital room under observation.
- Follow-up test confirmed heart condition → surgery required.
- Physician issued admission order → **patient status changed to Inpatient**.
- Patient underwent inpatient surgery and continued hospitalization.

Importance of Inpatient vs. Outpatient Status

1) Administrative and financial implications:
   - Status determines health insurance reimbursement.
2) Clinical implications:
   - Types and complexity of medical equipment differ by location:
     - Outpatient settings → simpler, diagnostic/treatment devices.
     - Inpatient settings → advanced life-support, surgical, monitoring equipment.

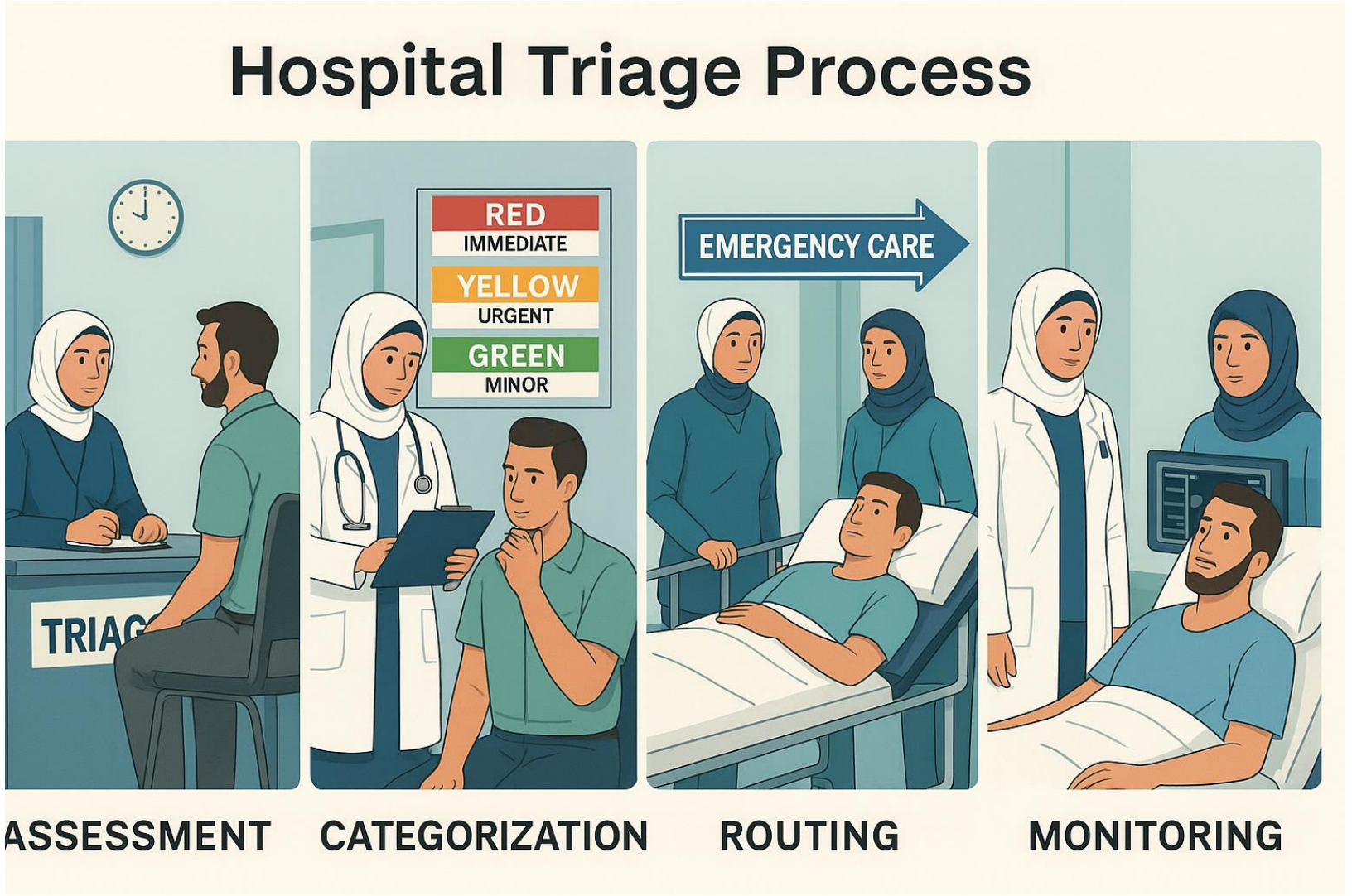
Classification of Hospital Departments: Inpatient and Outpatient Services

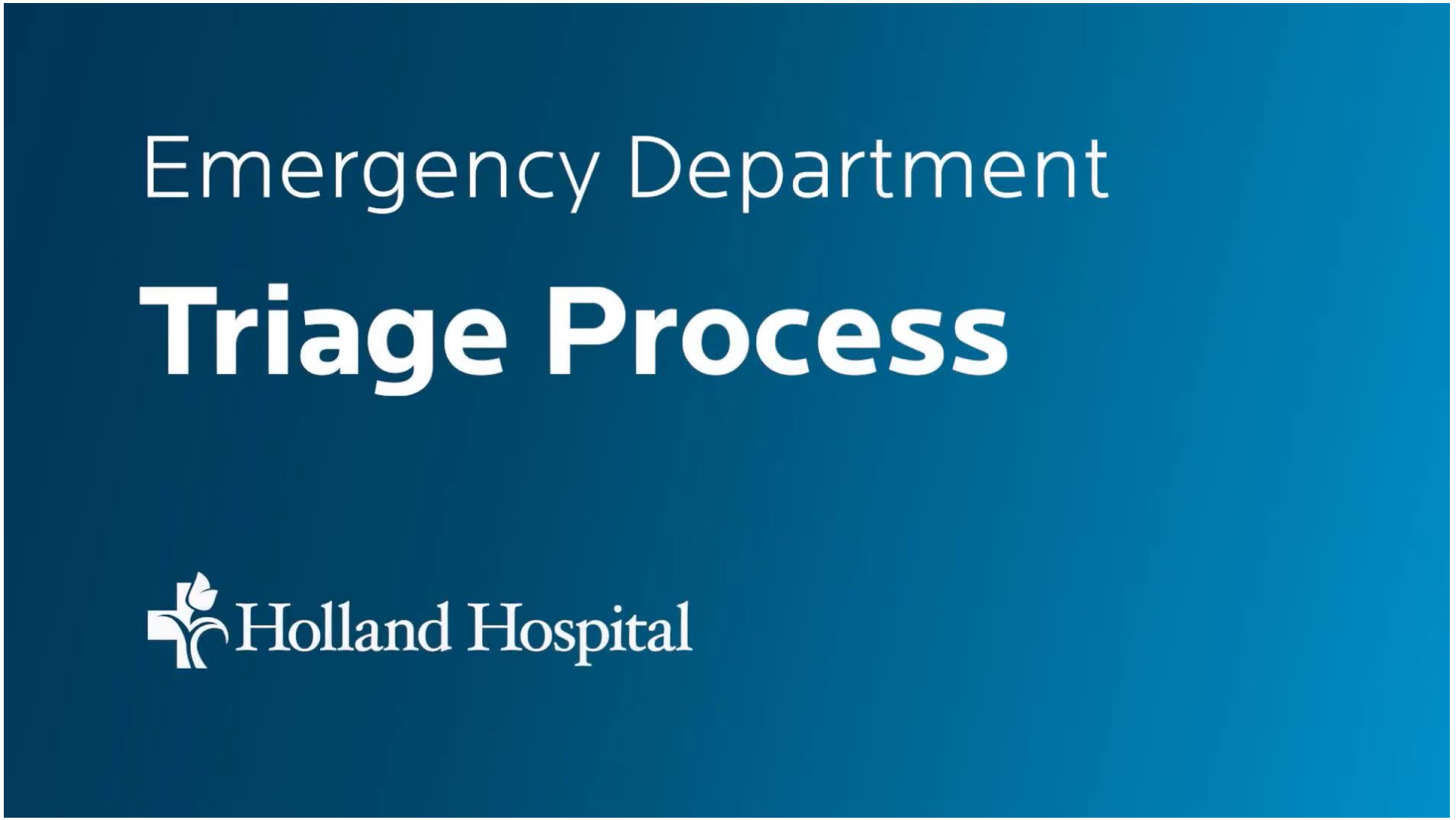


Triage in Hospitals (Patient Classification System)

- **Definition**: Triage is the process of quickly assessing and classifying patients based on the severity of their condition and urgency of treatment.
- **Purpose**: Ensures limited medical resources are directed first to those who need them most.
- **Common Levels**:
  - **Immediate (Red)** – Life-threatening, needs urgent care.
  - **Urgent (Yellow)** – Serious but not immediately life-threatening.
  - **Delayed (Green)** – Minor injuries, can wait.
  - **Expectant/Deceased (Black)** – Very low chance of survival or already deceased.
- **Use in Hospitals**: Emergency departments, disasters, and mass casualty incidents.
- So , Triage = Prioritizing patient care for maximum survival.

Triage in Hospitals





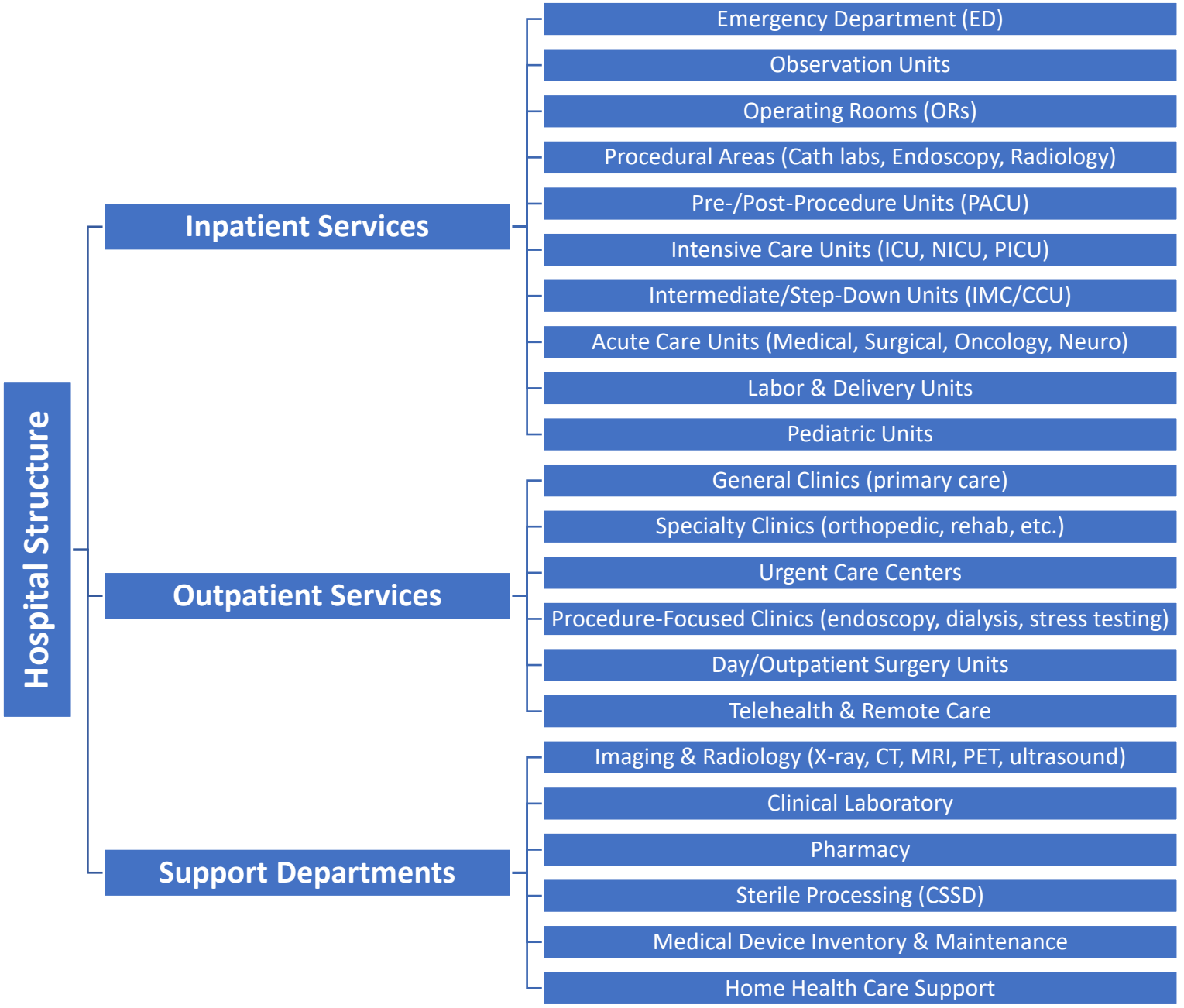
https://www.youtube.com/watch?v=cvbF3FP3RFk

Who Does Triage?

- **Usually done by trained medical professionals (Nurse / Doctor).**
- In **hospital emergency departments**:
  - **Emergency nurses** are often the **first to perform triage** (they are specifically trained to assess patients quickly).
  - In complex or unclear cases, **emergency medicine doctors** may step in.
- In **mass casualty / disaster medicine**:
  - Triage can be performed by **emergency physicians, trauma surgeons, anesthesiologists**, or specially trained **paramedics**.
- So, while **doctors may supervise**, triage is most commonly the responsibility of **emergency nurses and emergency medicine doctors**.

Relation to Inpatient & Outpatient

- **Triage is a *step before* deciding inpatient vs. outpatient.**
- After triage:
  - **Critical cases** (Red, some Yellow) → **Inpatient admission** (ICU, ER, wards).
  - **Mild cases** (Green) → Treated quickly and discharged as **outpatients**.
  - **Non-survivable / deceased** (Black) → Managed according to protocols (not admitted).
- **In short:**
  👉 Triage **decides urgency of care** → then hospital decides **whether patient stays (inpatient) or goes home (outpatient)**.



*Central Sterilization Supply Department

Outpatient Clinics care areas

Short treatment duration is the hallmark of clinic care.

- General primary care (basic, non-critical devices)
  - Scales, thermometers, exam tables, otoscopes/ophthalmoscopes, sphygmomanometers
- Specialty clinics
  - Orthopedic: cast saws, basic x-ray units
  - Rehab/Physical medicine: treadmills, bicycles, TENS, water baths
- Urgent Care
  - Manage non-life-threatening illness/injury when primary care is unavailable
- Devices: primary care equipment + labs (blood/urine), imaging (x-ray, simple ultrasound)
- Mostly non-critical, except emergency response equipment (e.g., defibrillators).

Inpatient patient care areas

Longer treatment duration with overnight admission and continuous monitoring.

- **General Wards:** Beds, IV pumps, vital signs monitors, oxygen systems; non-critical but escalatable.
- **ICU/NICU/PICU:** Life-threatening/unstable cases; ventilators, advanced monitors, infusion pumps, defibrillators; critical care.
- **Operating Rooms (ORs):** Major/minor surgeries; anesthesia machines, surgical lights/tables, electrosurgical units, endoscopic towers; high-resource critical area.
- **PACU/Recovery:** Post-surgery stabilization; bedside monitors, oxygen therapy, airway tools; short-term high-risk zone.
- **Specialized Units:**
  - Cardiac (CCU): Telemetry, defibrillators, EKG systems
  - Oncology: Chemo pumps, isolation equipment
  - Neurology: Neuro monitors, seizure precautions
- **Labor & Delivery & Postpartum:** Fetal monitors, delivery beds, infant warmers, neonatal incubators; routine + emergency-ready.

Note : Procedure-Focused Clinics

Equipment depends on the procedure performed , ex:

- **Stress test clinics:** treadmills, physiologic monitors
- **Endoscopy clinics:** GI scopes, bronchoscopes
  - Non-critical but may use anesthesia machines if sedation provided
- **Dialysis clinics:**
  - Dialysis machines → semipermeable membrane filters blood
  - Require complex mechanical/electrical systems for fluid management

Outpatient – 1 Emergency Departments (EDs)

Main Areas & Equipment

- **Triage:** stretchers, thermometers, BP cuffs, suction, ECG, IV pumps, vital monitors
- **Trauma/Critical:** x-ray, bedside ultrasound, crash carts, ventilators, intubation tools, warming devices, incubators
- **High-Risk Devices:** ventilators, defibrillators (require strict safety checks)
- **Patient Monitoring**
  - **ECG views:**
    - 3-electrode → **3 views** (basic)
    - 5-electrode → **6 views** (enhanced)
    - 12-lead (10 electrodes) → **12 views** *(diagnostic use)*
  - *Lead = physiological view, not just an electrical wire.*
  - **Pulse Oximetry:** SpO_2 + HR, cheap & common; **does NOT measure respiration rate**
  - **Automated BP Monitors:** detect arterial oscillations; often integrated with temp + SpO_2; faster/easier than manual method

Outpatient – 2 Observation Units

- Purpose: Short-term monitoring and treatment for patients before discharge or admission.
- Types of Units:
  - Low-acuity units: basic devices (beds, thermometers, BP monitors, scales).
  - Hospital-based units: beds/stretchers, air/oxygen regulators, thermometers, BP monitors, IV pumps, physiologic monitors, emergency carts with defibrillators.
- Medication & Fluid Delivery (IV Therapy):
  - Routes: oral, injection, transdermal, IV infusion.
  - IV pumps: very common devices, ensure accurate dosing, alarms for occlusion or completion.
  - PCA pumps: patient-controlled, lockable for controlled analgesia.
  - Smart pumps: drug libraries, interaction checks, dosage safeguards to reduce medication errors.

Outpatient – 3 Home Health Care

- Purpose: Provide medical care at home to reduce cost and improve patient outcomes.
- Equipment:
  - Basic devices: scales, BP monitors, pulse oximeters, glucose meters, nebulizers.
  - High-risk devices: ventilators, LVADs (left ventricular assist devices), infant phototherapy devices.
- Challenges:
  - Caregivers may lack technical training (device use, electricity, pneumatics, safety).
  - Home environment issues: unreliable power, variable temperature/humidity, hygiene problems.
  - Clinical engineers must plan preventive maintenance, safety checks, and emergency response procedures.

Outpatient – 4 Telehealth

- Purpose: Expands access to care for rural/remote patients and fills specialty gaps.
- Technology Use:
  - Home monitoring and diagnostic tools.
  - Examples: cardiac monitors, fetal monitors.
  - Remote monitoring of implanted devices (defibrillators, pacemakers).
- Requirements:
  - Secure and reliable internet connection.
  - Video cameras and data-transmission devices.
- Benefits:
  - Supports outpatient care continuity.
  - Reduces need for patient travel.

Inpatient – 1 Operating Rooms (ORs)

- Environment: Highly controlled, sterile spaces for surgical procedures; may be general or specialty-specific.

General OR Equipment:

- Operating table + overhead surgical lights (often foot-pedal controlled).
- Anesthesia machines deliver anesthetic gases (e.g., N_2O, sevoflurane, desflurane, isoflurane, halothane).
  - Functions: induce unconsciousness, paralysis, and pain control.
  - Require ventilators to support breathing during anesthesia.
  - Include capnography to monitor CO_2 exchange.
  - Use gas absorbers to remove anesthetic gases from breathing circuits.
- Depth of anesthesia monitored using EEG-based systems (scale ~0–100).

Surgical Technologies in ORs

- **Electrosurgical Units (ESUs):**
  - RF range 300–3000 kHz
  - Used for cutting and coagulation
  - Modes: monopolar (pencil + return electrode), bipolar (forceps/tweezers)
  - Produce smoke → smoke evacuators required.
- **LASERs:**
  - CO_2 LASERs for cutting/coagulation
  - Specialized: Nd:YAG, Argon
  - Used for tumors, tattoos, skin resurfacing.
- **Minimally invasive surgery (MIS):**
  - Scopes with fiber optics + video (laparoscopes, endoscopes, gastroscopes, bronchoscopes, cystoscopes, etc.)
  - Insufflators introduce CO_2 to create space in abdominal cavity.
- **Surgical microscopes:** binocular, mechanical, mounted on booms.
- **Surgical navigation & robotics:**
  - Real-time imaging integration
  - Robotics → joystick controls for precision.
- **Patient warming:**
  - Blanket warmers, IV/blood warmers, radiant warmers.

Inpatient – 2 Procedural Areas (Cath Labs, Endoscopy, Radiology)

- Purpose: Areas for specialized diagnostic or interventional procedures; similar to ORs but focused on specific interventions.
- General Equipment:
  - Procedure table, lights, anesthesia machine
  - Crash cart with emergency equipment
- Cardiac Catheterization Labs:
  - Single/multi-plane imaging systems
  - Patient cooling devices
  - Intravascular ultrasound (IVUS)
  - Contrast injectors
- Endoscopy Suites:
  - Endoscopes and processing/washer units
  - Ultrasound devices and video systems
- Other Procedural Areas:
  - Diagnostic & interventional radiology suites
  - Radiation oncology procedure areas

Inpatient – 3 - Pre- & Post-Procedure Areas (PACU)

- Pre-Procedure Rooms/Bays:
  - Temporary holding spaces before surgery/procedures.
  - High patient turnover, short stays.
- Post-Procedure / Recovery (PACU):
  - Monitors patients after surgery and anesthesia until stable for transfer.
- Typical Equipment:
  - Stretchers/beds
  - Physiologic monitors
  - Blanket warmers
  - IV pumps

Inpatient– 4 Intensive Care Units (ICUs)

- Purpose: Manage high-acuity patients requiring continuous monitoring and advanced life-support equipment.
- Types of ICUs:
  - Cardiac ICU
  - Neurology ICU
  - General ICU
- Cardiac ICU Equipment:
  - LVADs
  - ECMO (oxygenates blood + removes CO_2; similar to bypass)
  - Intra-aortic balloon pumps (reduce cardiac workload)
- Ventilators:
  - Support or fully replace breathing
  - Modes: preset rate, assist mode, apnea backup, high-frequency ventilation (500+ bpm)
- Neurology ICU:
  - Hypothermia/temperature management devices
  - Intracranial pressure (ICP) monitors
  - Ventilators
- General ICU Equipment:
  - Specialty beds
  - Dialysis machines
  - IV & feeding pumps
  - Wound vacs
  - Advanced physiologic monitors
  - Point-of-care analyzers
  - Patient lifts/ceiling booms

Inpatient – 5 - Intermediate / Step-Down Units (IMC / CCU)

- Purpose: For patients who require more care than standard wards but less than ICU-level support.
- Equipment:
  - Some critical-care devices: ventilators, dialysis machines.
  - Bedside physiologic monitors in each room.
  - Telemetry units for wireless cardiac monitoring while patients can walk/ambulate.
  - Includes standard acute-care equipment (IV pumps, beds, medication systems, etc.).

Inpatient – 6 - Acute Care Units (Medical / Surgical / Oncology / Neuro)

Purpose: Lower acuity than ICU/IMC; routine inpatient care.

Equipment:

- Portable vital signs devices (HR, BP, SpO_2)
- Sequential compression devices (prevent blood clots)
- IV pumps (very common devices)
  - PCA pumps → patient-controlled pain management
  - Syringe pumps → small-volume medication delivery
  - Feeding pumps → enteral nutrition (vs IV pumps for parenteral)

Inpatient – 7 - Labor & Delivery Units

- Purpose: Specialized units for childbirth with continuous monitoring of mother and fetus.
- Equipment:
  - Fetal heart monitors:
    - Doppler ultrasound (non-invasive)
    - Direct scalp electrode (used when fetal head is accessible)
  - Contraction monitoring via external transducer
  - Telemetry systems → allow wireless fetal monitoring while the mother walks/ambulates
- Post-Birth Safety:
  - Infant abduction prevention systems using RFID tags (on baby and sometimes mother)
  - Alarms activate if tag is removed or infant leaves authorized care area

Inpatient – 8 - Pediatric & Neonatal Units (NICU & PICU)

NICU Equipment (Neonatal Intensive Care):

- Incubators (isolettes): controlled temperature & humidity with access doors.
- Radiant warmers: open beds with overhead heater + skin sensor for feedback.
- Phototherapy: blue light (425–475 nm) to treat neonatal jaundice; fiber-optic blankets reduce side effects.

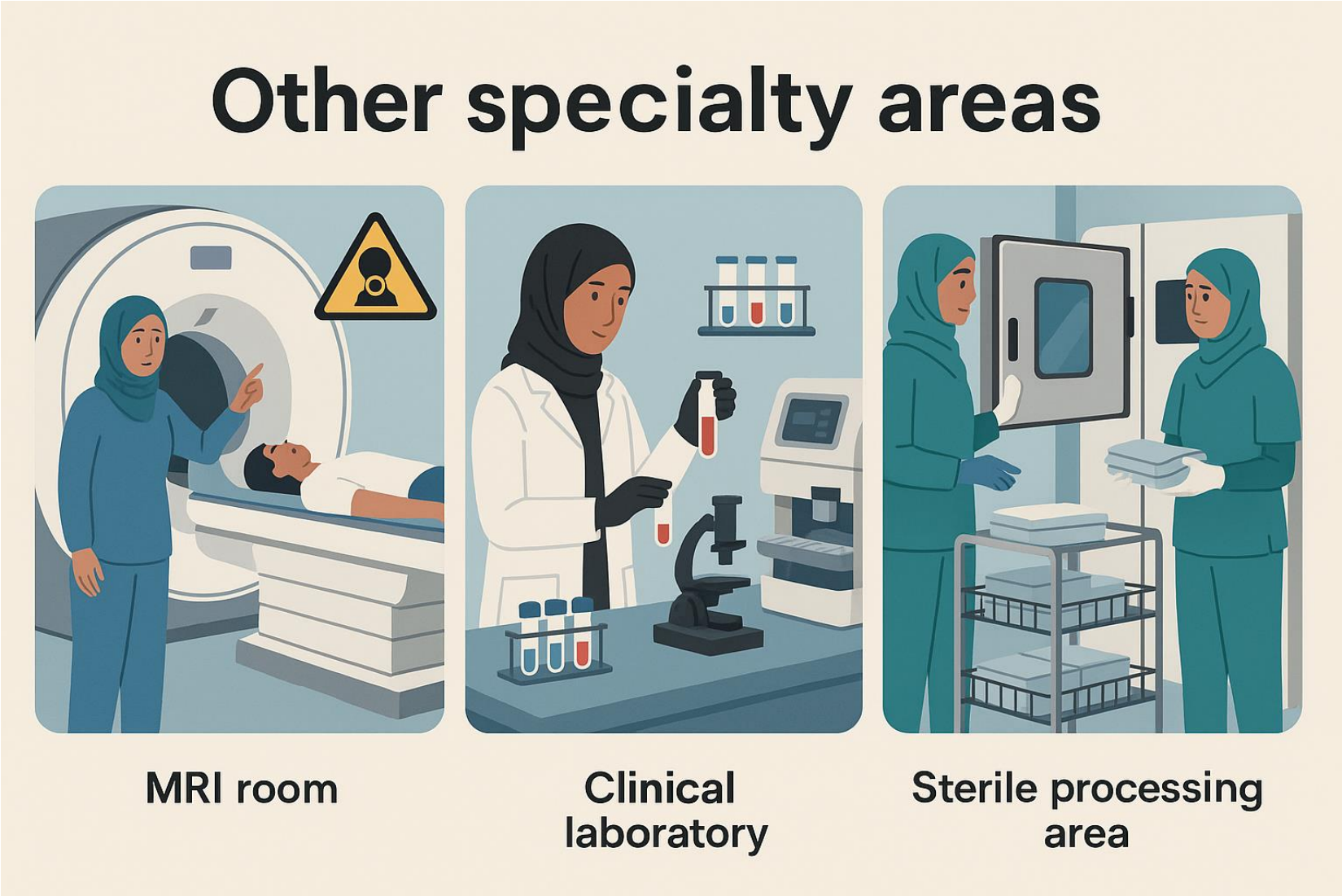
PICU Equipment (Pediatric Intensive Care):

- Specialty cribs and oxygen tents.
- Feeding pumps, breast pumps, milk warmers, solution warmers.
- Comfort-based equipment: play areas, consoles, toys for pediatric well-being.

Common Equipment Across Hospitals

- Headwalls:
  - Contain vacuum/suction regulators, air/oxygen flow meters, emergency outlets, gas blenders, and nurse call systems.
- Nurse Call Systems:
  - Range from basic indicator lights to fully integrated communication and paging systems.
- Hospital Beds:
  - Advanced beds support circulation, posture, airflow, and patient safety.
  - Features may include weight monitoring and exit alarms.
- Specialized Beds:
  - Orthopedic beds (assist bone healing and lifting support)
  - Bariatric beds (high weight capacity: ~600–1000 lbs)
  - Birthing beds (removable sections, support bars for delivery)

Other specialty areas



Imaging

- **Categories based on image generation method:**
  - **Ultrasound** – sound waves





Imaging

- **Categories based on image generation method:**
  - **Radiation-based** – X-rays, CT, PET, nuclear medicine, mammography









Imaging

- **Categories based on image generation method:**
  - **Magnetic field-based** – MRI



Core Medical Imaging Modalities

- Ensure safe operation, maintenance, calibration, radiation safety, and modality-specific environment setup (shielding, accessories, QC procedures).
- **Ultrasound**
  - Real-time soft tissue imaging using sound waves.
- **X-Ray & CT (Ionizing Imaging)**
  - X-ray → 2D structural imaging; CT → 3D axial scans using multi-detector systems.

Core Medical Imaging Modalities

- Ensure safe operation, maintenance, calibration, radiation safety, and modality-specific environment setup (shielding, accessories, QC procedures).
- **Fluoroscopy**
  - Continuous x-ray for real-time procedures (angiography, cath labs).
- **MRI (Magnetic Imaging)**
  - Uses magnetic fields + RF signals; shows detailed soft tissue + physiology.

Core Medical Imaging Modalities

- Ensure safe operation, maintenance, calibration, radiation safety, and modality-specific environment setup (shielding, accessories, QC procedures).
- **Mammography & DEXA**
  - Specialized x-ray systems: breast tissue imaging & bone density evaluation.
- **Nuclear Medicine (PET, SPECT)**
  - Uses radiotracers to image metabolism, blood flow, molecular activity.
  - PET/CT & SPECT/CT hybrid systems combine function + anatomy.

Core Medical Imaging Modalities

- Role in Clinical Engineering : Ensure safe operation, maintenance, calibration, radiation safety, and modality-specific environment setup (shielding, accessories, QC procedures).
- **Therapeutic Nuclear Applications**
  - Radiotherapy sources (I-131, brachytherapy seeds).
  - Support systems: sealed sources, LINAC-radionuclide interfaces, safety compliance.



Clinical Laboratory

- Analyzes patient samples for disease.
- Divisions: pathology, chemistry, hematology, genetics, microbiology, phlebotomy, blood bank.
- Equipment: microscopes, centrifuges, incubators, tissue prep devices; automated chemistry/hematology analyzers, specimen processors, apheresis machines.
- Systems: automation + LIS integration for direct result entry.

Pharmacy

- Prepares and dispenses drugs for patient populations.
- Limited equipment compared to other hospital areas.
- Technologies:
  - PCA pumps (pain management).
  - IV mixing robots in large pharmacies (require engineering support).
  - Automated dispensing systems: computer-controlled locked drawers/bins → restrict access & improve safety.

Sterile Processing

- Also called Central Sterile Services (CSSD) or Central Supply.
- Responsible for cleaning & sterilizing reusable devices and consumables.
- Equipment:
  - Scope washers
  - Sterilizers and autoclaves (high/low pressure, steam, ethylene oxide gas)
  - Goal: destroy germs, ensure safe reuse of equipment and consumables.

Sterile Processing (CSSD) – Medical Waste Incinerator

- Responsible for cleaning, disinfecting, and sterilizing reusable medical devices.
- Used for the final destruction of:
  - – Infectious medical waste
  - – Contaminated items that cannot be sterilized
  - – Pathology waste (tissues, organs, biological materials)
- May fall under CSSD administratively, but it is not part of the sterilization cycle.
- Part of hospital Infection Control & Waste Management systems.
- Medical waste treatment methods: **Incineration, Autoclave steam treatment, Microwave disinfection, Chemical treatment, Plasma pyrolysis → Final output: safe, non-infectious, reduced-volume waste ready for disposal.**

Think with me !? 🤔

- Why do you think hospitals classify devices as high-risk vs. non-high-risk instead of using FDA/MDR classes?(Hint: Think clinical use, not manufacturing.)
- How does a patient’s status (outpatient vs. inpatient) change the type of technology and monitoring they need?
- If you were a clinical engineer, which device category would you prioritize for preventive maintenance ， and why?(Diagnostic? Therapeutic? Life-support? Telehealth device?)
- In your opinion, which hospital department contains the most expensive medical equipment?


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Thank you

Abbreviation	Full Meaning
FDA	Food and Drug Administration
ICU	Intensive care unit
IMC	Intermediate care unit
MEMP	Medical equipment management plan
MI	Myocardial infarction
MRI	Magnetic resonance imaging
NFPA	National Fire Protection Association
NICU	Neonatal intensive care unit
PCA	Patient-controlled analgesia
PET	Positron emission tomography
PHI	Protected health information
PICU	Pediatric intensive care unit